

Christina Paolicelli

Embedded Software Engineer

CONTACT

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PROFESSIONAL EXPERIENCE

Senior Software Engineer, Algorithms | 2023 – Present

Shaper Tools – Handheld computer-vision-enabled CNC Router

- Designed and implemented computational geometry algorithms for automated pocket toolpath generation, including polygon offsetting, region decomposition, and robust handling of self-intersecting and nested contours.
- Improved high-speed cutting performance by optimizing toolpath generation strategies.
- Implemented computer vision-based marker detection for secure, unique pairing with a Bluetooth accessory.
- Modularized core geometry and vision systems into standalone libraries, enabling deterministic testing, improving reusability across embedded and WebAssembly targets.

Senior Software Engineer | 2021 – 2023

Rain Industries – Drones for Wildfire Response

- Lead software design and implementation for Fire Mapping. A system that identifies, locates, and suppresses wildfires from an autonomous helicopter.
- Stood up sensor test platform drone, coordinated test efforts, and wrote data analysis tooling.
- Operations crew member for autonomous helicopter test flight campaigns.

Senior Software Engineer | 2020 – 2021

Software Engineer | 2017 – 2020

BAE Systems

- Designed, developed, and tested software for embedded Electronics Warfare applications.
 - Developed automated testing infrastructure using Python and VBA.
 - Reverse engineered sensor systems signal acquisition processes.
 - Designed and developed proof of concept motor control and communications software.
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EDUCATION

Johns Hopkins University | Grad 2022

Master of Science in Computer and Electrical Engineering

Thesis: Deep Learning Framework for Character Recognition in Low Quality License Plate Images

Rensselaer Polytechnic Institute | Grad 2017

Bachelor of Science in Electrical Engineering

SKILLS

C++

Python

Git

INTERESTS

Figure Skating

Snowboarding

Knitting